

VARUNA06 — Research & Technical Validation

Objective:

Validate the scientific basis of multi-sensor seafloor exploration and anomaly detection using established marine datasets and peer-reviewed technical literature.

Data Sources:

- **GEBCO:**
Provides global bathymetric terrain data for seabed depth and topographic context.
- **NOAA NCEI:**
Provides marine geophysical datasets including bathymetry, magnetics, gravity, seismic and side-scan/subbottom information.
- **EMODnet Geology:**
Provides harmonised geological, sediment and marine-mineral information.
- **International Seabed Authority:**
Provides information on exploration areas and contracts for polymetallic nodules, polymetallic sulphides and cobalt-rich ferromanganese crusts.

Technical Literature:

- **Magnetic anomaly detection:** IEEE Sensors Journal review.
- **Electromagnetic seabed sensing:** IEEE underwater controlled-source electromagnetic sensing study.
- **AI seabed classification:** IEEE deep-learning seabed sediment classification study.
- **Acoustic + magnetic fusion:** IEEE deep-sea seafloor classification study.
- **Optical + acoustic fusion:** IEEE underwater data-fusion study.